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Water balance in an Atlantic Forest remnant: focus on representative tree species --Manuscript Draft--

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Abstract:	Carlos R. de Mello, Ph.D.
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	Kelly Tonello, Ph.D. Associate Professor, Federal University of Sao Carlos kellytonello@ufscar.br Dr. Tonello has developed forest hydrology studies in tropical forest in Brazil and has conducted a good research.
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October 31, 2024

Dear Editor:

I am writing to submit the manuscript “**Water balance in an Atlantic Forest remnant: focus on representative tree species**” to the Editorial Board of *Ecohydrology and Hydrobiology* for publication.

This study is the first conducted in the Atlantic Forest in Brazil in which the tree species were analyzed in plots, considering the root and above-ground biomass and soil moisture observed throughout the soil profile down 1.80 m. We study the dry period of the 2023/2024 hydrological year to evaluate how the trees deal with the water fluxes related to root distribution and biomass. Our results have great potential for application in studies related to ecosystem services that this very sensitive biome in Brazil offers.

I will be the corresponding author and can be reached by e-mail at crmello@ufla.br.

Thank you very much for your disposition.

Best wishes



Prof. Carlos R. de Mello - Corresponding author

Water balance in an Atlantic Forest remnant: focus on representative tree species

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Water balance in an Atlantic Forest remnant: focus on representative tree species

Abstract

The Atlantic Forest has faced deforestation and prolonged droughts, affecting the ecosystem service. This study evaluates the water balance focused on soil moisture and forest species in a Montane Semideciduous Seasonal Forest (SSMF) remnant. Soil moisture, gross precipitation (GR), canopy interception (CI), effective precipitation (EP), above-ground biomass (AGB), root biomass (BR), and soil hydrological properties along the profile down to 1.80 m were assessed in the dry period of 2023/2024. EP, AGB, and BR influenced ET and soil moisture dynamics. The highest EP values were observed in *Miconia willdenowii*, while the lowest were in *Xylopia brasiliensis*. The highest and lowest CI were observed in *Xylopia brasiliensis*. Root zone water storage showed a declining trend, with the highest values in *Miconia willdenowii*. ET has been critical in April, July, and September, with *Miconia willdenowii* and *Copaifera langsdorffii* presenting the highest values, and AGB strongly correlated with CI and ET. The dynamic nature of the ecosystem is evident in the temporal variations (CVt) of soil moisture, influenced by rainfall and root uptake. The greatest spatial variability was observed in the superficial layer (0-20 cm), stabilizing with the depth. The Temporal Stability Index (TSI) indicated greater stability at *Blepharocalyx salicifolius* plot. This study highlights the critical importance of soil water storage and ET in a tropical forest ecosystem, especially under drought conditions, allowing the identification of species that can be more successfully used to recover degraded areas.

Keywords: soil moisture, evapotranspiration, dry period, Semideciduous Atlantic Forest, tree species

24 Introduction

25 The water balance in forest formations is not just a significant factor but a crucial one in
26 understanding the role of vegetation in the soil-vegetation-atmosphere continuum. This connection is
27 maintained through a complex, non-static process with elasticity and a robust network of interactions (Han
28 et al., 2024). Understanding the water balance is of utmost importance, as forest formations provide many
29 ecosystem services, particularly in regulating the hydrological cycle, reducing direct surface runoff,
30 increasing percolation and groundwater recharge, and nutrient cycling (Pizarro et al., 2022).

31 One of the most relevant forest formations in Brazil is the Atlantic Forest, recognized as a
32 significant biosphere reserve on the planet due to its water yield capacity and diversity of endemic species
33 (Mello et al., 2019). This biome has been essential for maintaining hydrological flows during the dry season
34 and regulating maximum flow extremes during the wet season in rivers (Junqueira Júnior et al., 2019). In
35 Southeastern Brazil, the Atlantic Forest has faced prolonged drought periods since 2014 (Rodrigues et al.,
36 2021b), which threatens the fragile balance of this ecosystem. In this region, the Montane Seasonal
37 Semideciduous Forest (MSSF) predominates and is established in areas at altitudes over 500m.

38 Several studies have been carried out in remnants of the MSSF in Brazil to understand the spatial
39 and temporal behavior of soil moisture (Junqueira Júnior et al., 2017; Oliveira et al., 2021; Rodrigues et al.,
40 2020), the influence of vegetation on water inputs through canopy interception (Junqueira Júnior et al.,
41 2019; Rodrigues et al., 2021a) and evapotranspiration (Guauque-Mellado et al., 2022) under drought
42 conditions. However, few studies have associated the behavior of soil moisture, the root concentration of
43 forest species, and the water balance conducted over representative species. These aspects are fundamental
44 for understanding ecosystem services.

45 MSSF is home to a variety of tree species, 50% of which are deciduous, i.e., losing their leaves and
46 absorbing water from deeper layers to cope with prolonged drought periods (Rodrigues et al., 2021b,
47 Mantovani et al., 2022). Therefore, deepening our understanding of how water storage varies in the soil

profile and its relationship with the concentration of species' root biomass and the soil's hydrological properties is essential. In addition, it is necessary to consider other interrelated variables, such as effective precipitation (EP; a rainfall that hits the forest ground), aboveground biomass (AGB), root biomass (BR), and canopy interception (CI) to understand how much water trees lose through evapotranspiration (ET), especially in the dry season. This would improve our knowledge of how soil, water, and tree species connect in different water availability conditions, shedding light on the species' adaptability and the role played by the soil in dry conditions.

This study aims to answer the question: How important is the distribution of roots in the soil profile and the amount of above-ground biomass for the resilience of representative Atlantic Forest species to droughts? This will allow us to advance our understanding of the relationships between hydrological variables and vegetative characteristics below and above ground in water dynamics, especially during the dry season in an MSSF.

Thus, the objectives of this work are i) to determine the water balance of representative species of the Atlantic Forest; ii) to evaluate the influence of hydrological variables on the dynamics of evapotranspiration; iii) to investigate the relationship between the root biomass concentration, soil moisture, and soil hydrological properties at different depths; and iv) to analyze the spatial and temporal variation of soil moisture and the stability of soil water storage considering a 180 cm soil profile.

Material and Methods

Location and general aspects of the study area

The study area is in southeastern Brazil, in Minas Gerais state, at coordinates 21°13'40"S and 44°57'50"W, with an average altitude of 925 m (Junqueira Junior et al., 2017). It is a forest remnant classified as a Montane Semideciduous Stationary Forest (MSSF), one of the formations of the Atlantic Forest (Junqueira Júnior et al., 2017; Mantovani et al., 2022).

The remnant occupies an area of 6.35 ha and is surrounded by agroforestry and urban areas (Figure 1). The terrain is gently undulating, with slopes ranging from 5 to 15% (Junqueira Junior et al., 2017). The soil is classified as a dystrophic red Latosol (Junqueira Junior et al., 2017), with a granular structure and high porosity that provides high water infiltration capacity (Guauque-Mellado et al., 2022).

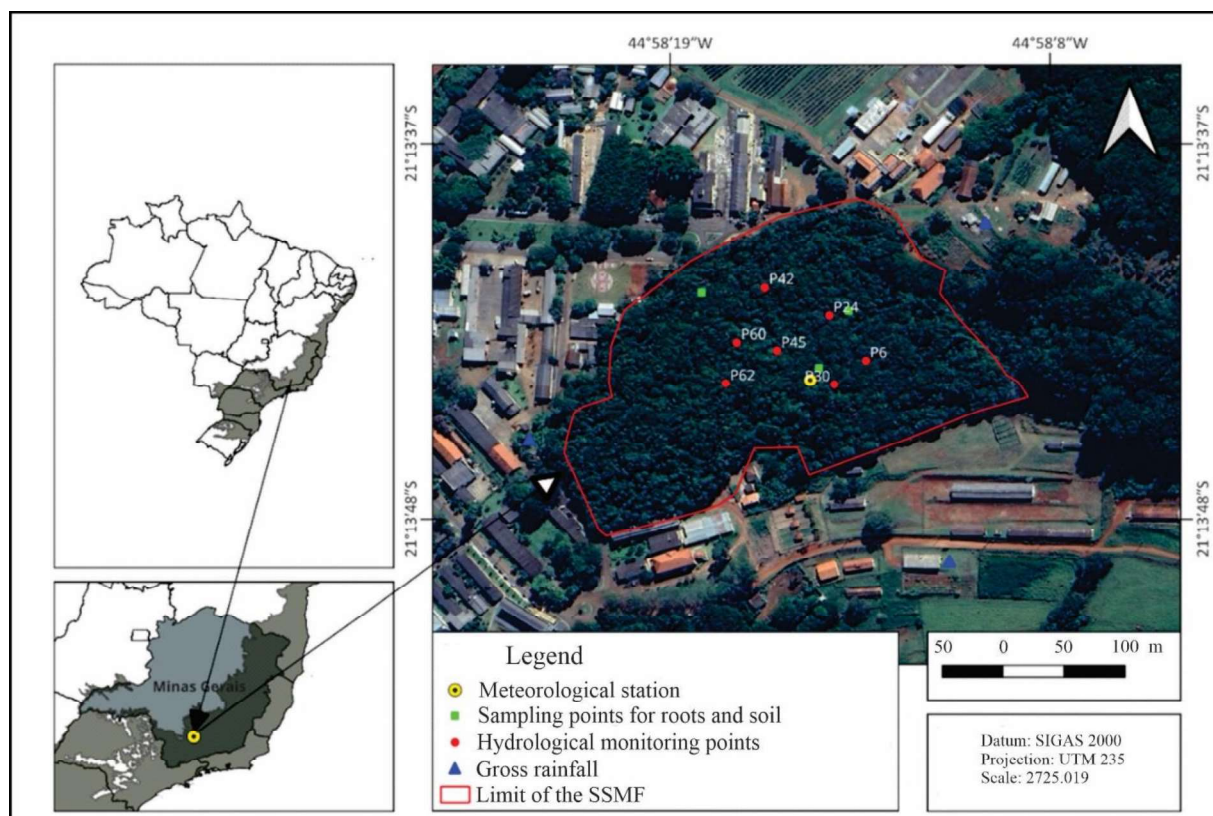


Figure 1. Geographical location of the MSSF, weather station, and monitoring plots for gross rainfall (GP), throughfall (TF), stemflow (SF), soil moisture, and soil sampling for root biomass characterization.

The region's Köppen climate classification is Cwa, a subtropical climate with rainfall concentrated in the summer (December to March) and well-defined seasons (Mantovani et al., 2022). The average annual rainfall is 1,383.4 mm, of which 1,186 mm (85.7%) occurs between October and March (rainy season) and 197.4 mm (14.3%) between April and September (dry season). The average annual potential evapotranspiration is 1,286.6 mm, ranging from 292.4 mm in January to 9.5 mm in July. The average annual

temperature is 20.6°C, ranging from 15.7°C in July to 27.5°C in February, and the average annual air humidity is 71.0% (INMET, 2020).

The MSSF has a semi-deciduous characteristic, i.e., 50% of the trees lose their leaves in the dry season (Mantovani et al., 2022; Rodrigues et al., 2021b). The last forest inventory, carried out in 2017, found a tree density of 1,021 trees/ha and a basal area of 120.85 m² ha⁻¹ (Junqueira Júnior et al., 2019). The most abundant species are *Copaifera langsdorffii* (Desf) (Fabaceae) (14.3%), *Xylopia brasiliensis* (Spreng) (Annonaceae) (13.7%) and *Miconia willdenowii* (Klotzsch ex Naudin) (Melastomataceae) (5.5%), representing 30% of the forest cover (Junqueira Júnior et al., 2019; Mantovani et al., 2022; Mantovani et al., 2023). These species have a higher population density in southeastern Brazil, demonstrating their importance on local and regional scales (Mantovani et al., 2022).

Hydrological monitoring

Since 2012, hydrological monitoring has been carried out at MSSF, focusing on weather conditions above and below the canopy, gross precipitation, throughfall, stemflow, and soil moisture in the 0 - 100 cm layer. Two Hobbo weather stations are installed on the site, one below and the other placed above the canopy on the top of a 22-m high meteorological tower to monitor precipitation, wind speed and direction, atmospheric pressure, net radiation, and air temperature/relative humidity at hourly intervals.

From the forest inventory, plots measuring 12 m x 50 m were selected for hydrological monitoring. The representative tree species were located within these plots to install the monitoring set, composed of a Ville de Paris rain gauge, an apparatus for stemflow collection, and an access tube for soil moisture measurement. The rain gauge, with an open area of ~ 400 cm², was installed 150 cm above the forest floor to avoid splash-in. The stemflow apparatus is an open hose spirally nailed to the tree trunk, draining the running water into a collection container (~ 30 L). The TECANAT PC access tube was installed to measure soil moisture up to a depth of 1.80 m using the TRIME-PICO (IMKO GmbH) sensor. Next to the monitoring

trees, root sampling for biomass assessment was carried out (Figure 1). Table 1 shows the representative species of these monitoring plots and their dimensions used for the water balance.

Table 1. Representative forest species aspects used in this study.

Plot number	Scientific Name	Popular name in Brazil	Height (m)	DBH* (cm)
42	<i>Blepharocalyx salicifolius</i> (Kunth)	Murta	12.0	19.50
60	<i>Xylopia brasiliensis</i> (Spreng.)	Pindaíba	15.0	17.00
62	<i>Xylopia brasiliensis</i> (Spreng.)	Pindaíba	13.0	18.80
45	<i>Copaifera langsdorffii</i> (Desf.)	Copaíba	16.4	32.10
30	<i>Miconia willdenowii</i> (Klotzsch ex Naudin)	Miconia prateada	12.5	23.10
6	<i>Xylopia brasiliensis</i> (Spreng.)	Pindaíba	15.4	35.00
24	<i>Miconia willdenowii</i> (Klotzsch ex Naudin)	Miconia prateada	13.0	25.00

*Diameter at Breast High

Precipitation was collected at least four hours after each rainfall event, at least four hours after the end of precipitation or in the following morning for those events in the late afternoon to ensure that the canopy dried out and avoid overlapping events (Mantovani et al., 2022).

The effective precipitation (EP, mm), which is the amount of rainfall that reaches the soil, is calculated based on stemflow (SF, mm) and throughfall (TF, mm) in each monitoring plot:

$$EP = TF + SF \quad (1)$$

Canopy rainfall interception (CI) was determined as follows:

$$CI = GP - EP \quad (2)$$

Soil moisture, soil water storage, and water balance

Soil moisture data (θ) were collected every two days during the dry season (April - September) of the 2022/2023 hydrological year in the depths of 20, 40, 60, 100, 140 and 180 cm. The monitoring depth covers most of the effective root zone for water absorption since, in the root sampling campaign, we observed that fine roots exist primarily up to 180 cm in the soil profile. The soil water storage (SWS) and its consecutive variation were calculated as follows:

$$SWS = \sum_{i=1}^n \left(\frac{(\theta_i + \theta_{i+1})}{2} \right) * h \quad (4)$$

$$\Delta W = SWS_{(t)} - SWS_{(t-1)} \quad (5)$$

Where i is the soil moisture monitoring depth, n is the number of depths (i.e., 20, 40, 60, 100, 140 and 180 cm), θ is the soil moisture ($\text{cm}^3.\text{cm}^{-3}$), h is the thickness of the layer between i and $i+1$ (mm), $SWS(t)$ is the storage measured at time t and $SWS(t-1)$ is the storage measured at time $t-1$.

The water balance was conducted for each specie during the dry season because deep percolation (below 180 cm) is negligible in this period. Thus, the soil water balance response variable only consists of evapotranspiration (ET_{WB}):

$$\frac{ET_{WB}}{\Delta t} = \frac{EP - \Delta W + CI}{\Delta t} \quad (6)$$

Quantification of root biomass in the soil profile

The monolith sampling method was used to quantify root biomass (RB) (Ratuchne et al., 2016). This method is the most suitable in areas without information on biomass (e.g., native forests) and consists of sampling soil blocks (monoliths). In this method, three monoliths were excavated close to the hydrological monitoring points (Figure 2a). The monoliths extracted had dimensions of 25 cm x 25 cm x 180 cm, divided into six geometric profiles (layers): 0 - 20 cm, 20 - 40 cm, 40 - 60 cm, 60 - 100 cm, 100 - 140 cm and 140 - 180 cm (Figure 2a).



Figure 2. Root sampling by monoliths inside the MSSF (a) and root diameter categories obtained in each layer sampled (b).

The soil and collected roots were weighed, and the most detachable roots were separated by hand. The soil was then submerged in water to separate the remaining and less detachable roots, using a set of three overlapping sieves: the top one with a 5 mm mesh, the middle one with 3 mm, and the bottom one with 1 mm (Ramos et al., 2018; Ratuchne et al., 2016; Ribeiro et al., 2024b).

Three categories were considered to classify the roots by diameter (Ratuchne et al., 2016): fine roots (< 2 mm), fine intermediate roots (2 - 5 mm), thick-intermediate roots (5 - 50 mm) and thick roots (> 50 mm) (Figure 2b).

Estimating above-ground biomass

To estimate the aboveground biomass (AGB) of the representative trees, the Pantropical model (Chave et al., 2014) was used, as it performs better in tropical forests and bioclimatic conditions, including the Brazilian Atlantic Forest. Measurements of diameter at breast height—DHB (cm) and tree height—H (m) were taken to capture variations in volume between trees.

$$AGB_{est} = 0.0673 * (\rho D^2 H)^{0.976} \quad (7)$$

Where ρ (g cm^{-3}) is the wood density used to convert volume into dry mass, D is the diameter at breast height – DBH (cm), and H is the trees' height (cm).

Characterization of the soil physical attributes

Soil samples were collected at the same locations as the root samples, considering the same layers, i.e., 0 - 20 cm, 20 - 40 cm, 40 - 60 cm, 60 - 100 cm, 100 - 140 cm, and 140 - 180 cm, for the determination of soil saturated hydraulic conductivity (K_{sat}), particle density (D_p), bulk density (BD), and porosity (PT) in laboratory conditions.

Data analysis

Analysis of variance and Tukey's test ($\text{prob} < 0.05$) were carried out to compare the root distribution between the soil layers according to their categories. The Randomized Complete Block Design was used in the experimental arrangement of subdivided plots, where the layer was considered the main factor and the root categories as the subfactor.

To assess the relationship between soil moisture and root biomass (RB) in the soil profile, a correlation analysis was carried out using Pearson's coefficient based on their average values, considering the nearby sampling points for root biomass and soil moisture monitoring. According to Torres et al. (2020), the correlation intervals can be interpreted as follows: 0.9 to 1.0 (-0.9 to -1.0) indicates a “very high” positive (negative) correlation; 0.7 to 0.9 (-0.7 to -0.9) represents a “high” positive (negative) correlation; 0.5 to 0.7 (-0.5 to -0.7) a “moderate” positive (negative) correlation; 0.3 to 0.5 (-0.3 to -0.5), a “weak” positive (negative) correlation; and 0.0 to 0.3 (0.0 to -0.3) represents a “very weak” positive (negative) correlation.

Results and Discussion

Distribution of root biomass

Figure 3 illustrates the distribution of roots by diameter category in each layer considered in this study. The averages of the three profiles collected inside the MSSF between July and August 2023 were considered.

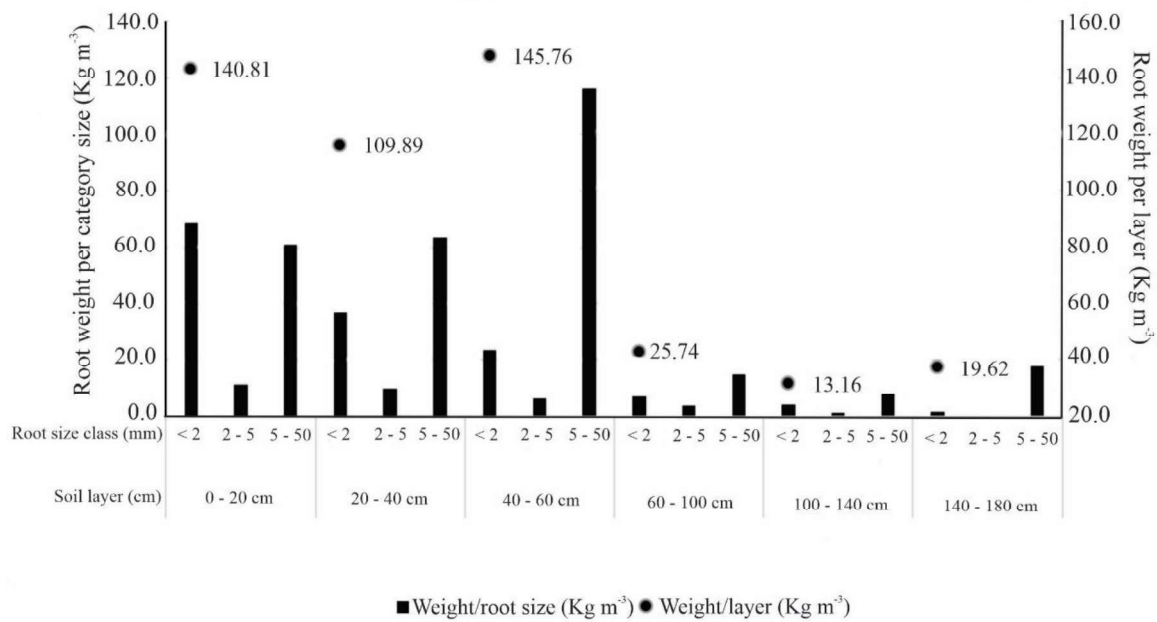


Figure 3. Root mass (Kg/m³) in each diameter category (mm) and total root mass (Kg/m³) at each soil depth obtained for the MSSF.

Regarding the difference in diameter categories throughout the layers, the variance analysis showed no significant interaction between the root sampling layers and the diameter categories, i.e., the concentration of roots per category does not necessarily depend on the layer in the soil profile. On the other hand, the same analysis showed significance ($p < 0.05$) between diameter categories in the same layer, and the Tukey test revealed a statistical difference between the means of these categories. This difference indicates significant variations in the distribution of roots in different diameters.

A clear vertical distribution of root biomass can be seen, with the highest concentration in the 20 – 60 cm layer and a subsequent decrease with depth. The greatest root biomass was concentrated in the 0 – 20 cm, 20 – 40 cm and 40 – 60 cm layers, with masses of over 100 kg m⁻³. The root masses in deeper layers (60 – 100 cm, 100 – 140 cm, and 140 – 180 cm) showed values less than 30 kg m⁻³. Similar behavior in the distribution of root biomass was observed by Luo et al. (2019) in deciduous and coniferous forests in China, characterized by a monsoon climate. This was expected since most water and nutrient uptake happens in the shallow layers (Basílio et al., 2024; Leonova et al., 2022; Ratuchne et al., 2016).

By distributing the roots according to diameter category (Ratuchne et al., 2016), the largest root weights were found in the category of thick intermediate roots (5 - 50 mm) in all layers, except 0 - 20 cm, where the thin roots (< 2 mm) stood out. The category of fine intermediate roots (2 - 5 mm) showed lower values in all layers. The finest roots (< 2 mm) are responsible for absorbing water and cycling carbon and nutrients from the litter decomposition (Basílio et al., 2024; Leonova et al., 2022; Ratuchne et al., 2016), and these roots are generally concentrated in the surface soil layers (Kulmann et al., 2022). On the other hand, the thicker roots are responsible for supporting and anchoring the plant to the soil, conducting solutes, and the stability and expansion of the system, and they are present in the deeper layers (Morais et al., 2017; Ratuchne et al., 2016; Ribeiro et al., 2024b). Generally, thick roots contribute less to nutrient cycling and water absorption than fine roots (King et al., 2021). However, trees of different sizes and ages have different root distributions. The larger ones are responsible for greater biomass concentration even in deeper layers, easing water uptake compared to the shorter ones that rely mainly on the shallow layers (Demir et al., 2024). The remnant heterogeneity and seasonal characteristics partly explain the distribution of roots up to the 160-180 cm layer. The development of fine roots with depths may indicate the adaptation of larger trees to the wetting-drying cycles to uptake water from deeper layers in the dry period (Demir et al., 2024), increasing the remnant resilience.

Previous studies in different ecosystems did not indicate pattern results. However, many of them have focused on the finer roots, which tend to be representative of forest ecosystems. For example, Lima et

221 al. (2011) found an average of 591 g m^{-2} , equivalent to 2.96 Kg m^{-3} of fine root biomass in forest fragments
222 in the dry season, considering the 0 – 20 cm layer. Therefore, the results of this study indicate that the
223 remnants of the SSMF concentrate root biomass mainly in the finer roots and surface layers.

224 *Water balance*

225 The series illustrated in Figure 4, segmented by month, shows the temporal dynamics of the water
226 balance elements (EP, SWS, and CI) and ΔW from April to September 2023 at each plot, considering the
227 representative tree species in the SSMF remnant.



Figure 4. Monthly time series of water balance elements: effective precipitation (EP), soil water storage (SWS), canopy interception (CI), and variation in soil water storage (ΔW) in each representative species of the SSMF during the period from April to September 2023.

During the monitoring period, gross rainfall (GR) was 250.7 mm, with the highest values recorded in April, August, and September, ranging from 23.1 mm to 45.3 mm, 7.5 mm to 23.9 mm and 36.9 mm to

50.5 mm, respectively, at all the monitoring plots. The average accumulated effective rainfall (EP) was approximately 188.2 mm over the period, resulting from 10 rain events. This average EP is close to that observed by Junqueira Júnior et al. (2019) in the same remnant but using 32 observation plots during the dry periods of the 2012/13 and 2013/14 hydrological years, which were 183 mm and 189 mm, respectively.

Canopy interception (CI) followed the temporal variability of EP, with an average of approximately 50 mm. This value corresponds to approximately 22% of GR. The average CI was higher than that of Junqueira Júnior et al. (2019) for the dry period, possibly due to the lower rainfall intensity and the smaller number of precipitation events observed throughout the analysis period of this study.

The months with the highest GR (April, August, and September) are responsible for the highest EP and CI values (Figure 4), as well as the greatest variability (Figure 5). Considering all the monitoring plots, EP ranged from 3.2 to 45.1 mm in April, from 5.4 to 23.0 mm in August, and from 23.5 to 50.2 mm in September. Following the behavior of EP (Figure 5), CI reached 17.6 mm in April (*M. willdenowi*), 14.4 mm in August (*M. willdenowi*) and 20.3 mm in September (*X. brasilienses*). On the other hand, the critical months for EP were May, June, and July due to lower GP (Figure 5). In June, an average of 4.08 mm was observed for EP and 1.3 mm for CI. This behavior of EP and CI in the period is related to the intra-annual variability of rainfall, as May, June, July, and August are historically the drier months in the study region (Guauque-Mellado et al. 2022; Junqueira Júnior et al. 2017; Oliveira et al. 2021).

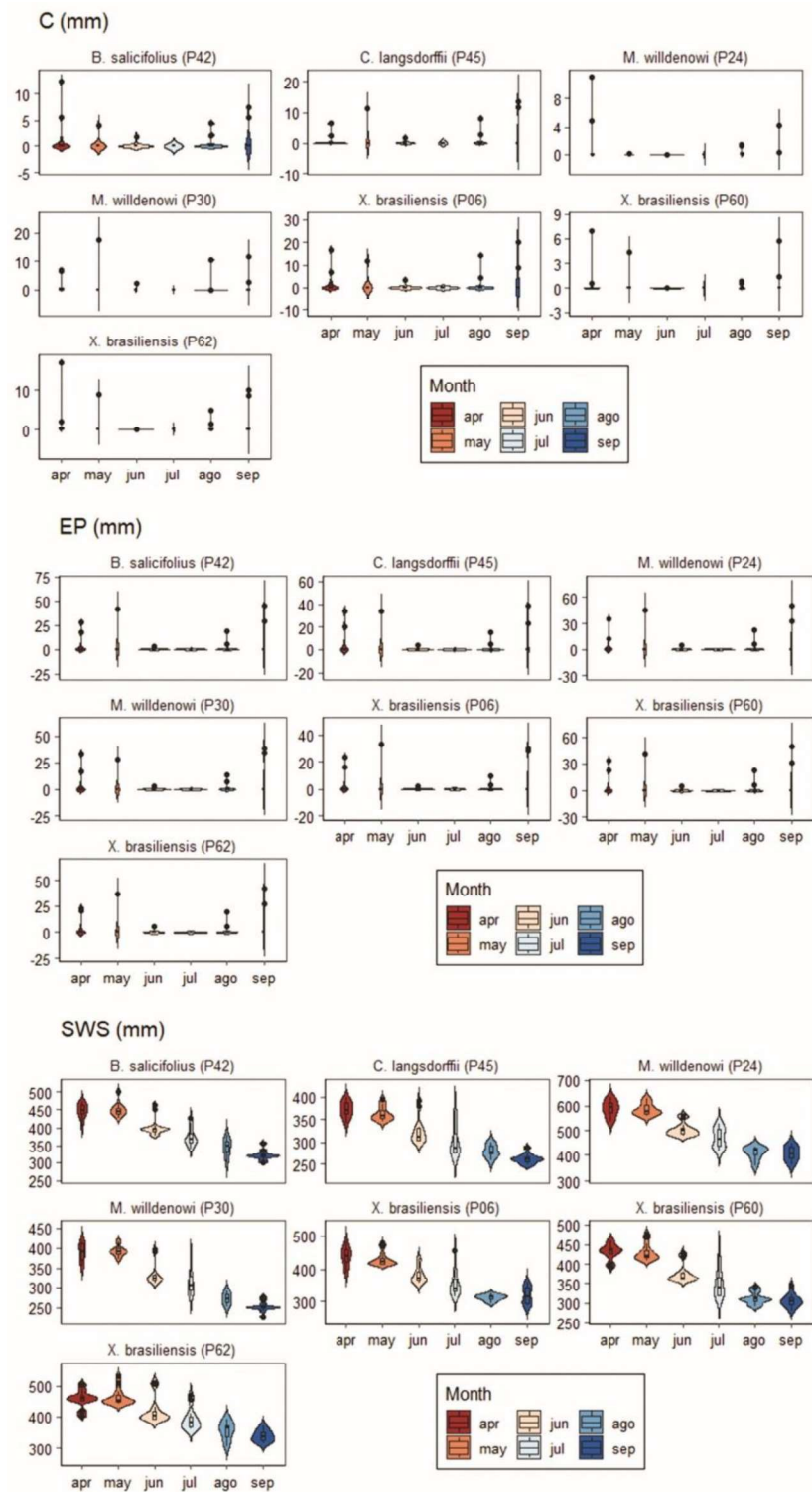


Figure 5. Water balance elements (EP, SWS, and CI) at each point with the representative species during the study period (dry period) at SSMF.

The representative plots of *Xylopia brasiliensis* presented a monthly EP of up to 49.2 mm (in plot 60), while the maximum CI was 20.3 mm (plot 2). In the tree of plot 6 (*X. brasiliensis*), EP values were consistently lower than those found in other species (from 2.2 mm to 30.2 mm). On the other hand, *M. argyrophylla* presented the highest EP values (from 5.3 mm to 50.2 mm), surpassing all the trees evaluated, with a maximum CI of 11.1 mm. In plot 30 (*M. willdenowii*), EP varied from 3.3 mm to 38.8 mm, with a maximum CI of 17.6 mm. The other trees, *B. salicifolius* and *C. langsdorffii*, presented EP ranging from 3.5 mm to 44.9 mm and 3.6 mm to 38.6 mm, respectively, with maximum CI values of 12.4 mm and 13.4 mm, respectively.

The species showed irregular behavior concerning EP and CI influenced by the canopy's architecture and the forest's specific characteristics at each plot (Lian et al., 2022; Mello et al., 2024; Oliveira et al., 2024). In some plots, especially where *B. salicifolius* and *X. brasiliensis* are present, other species were observed overlapping the canopy of these trees, influencing mainly the CI pattern.

The SWS at all monitoring plots showed higher values at the beginning of the series (April), with a general downward trend throughout the evaluation period and fluctuations due to the rain events (Figure 6). We observed that $EP \geq 4$ mm was the threshold for a more significant response in soil water storage. Liu et al. (2020) also observed a response of soil moisture to precipitation in the range of 3.5 to 5.6 mm in a native tropical forest. However, the SWS time series (mm) reveals a general reduction trend throughout the evaluation period (Figure 6). The plots representing *X. Brasiliensis* (6, 60, 62) showed this behavior, and as this specie is dominant in SSMF, SWS overall reduction can be linked to the influence of evapotranspiration. This is based on the temporal average soil moisture behavior, with a downward trend up to the depth of 140 cm (Figure 6). Higher SWS values were observed from this depth, meaning the trees uptake water up to 140 cm. In other species, SWS continuously increases from a depth of 60 cm, and roots uptake water concentrated in the shallow layers. These differences may be due to the spatial variability of soil properties (mainly those related to water retention) and species characteristics. Besides trees' characteristics of water uptake, soil water availability also drives root distribution. The greater SWS does

not directly translate into greater availability because water retention relies on the distribution of pore sizes (macro and micropores). A higher concentration of micropores may hamper the water uptake, forcing trees to seek water in deeper layers (Demir et al., 2024).

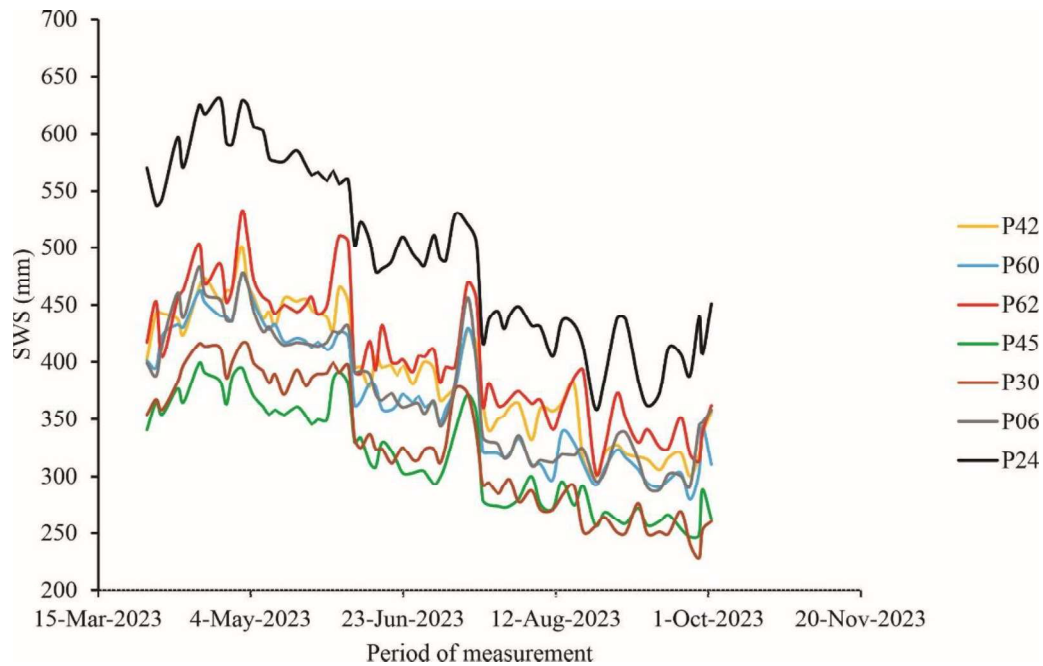


Figure 6. Temporal behavior of SWS (mm) in the soil profile during the study period (dry period) in the SSMF.

July presented the greatest variability in SWS, mainly in plots 45, 30, 6, and 60, although no EP was observed in this month. Rodrigues et al. (2021b), in the same remnant, found the presence of discrepant values in June, July, and August 2015. The explanation for this behavior is based on the fact that few rainfalls occur in this period, and the Dystrophic Red Oxisol has a high permeability. This way, the water redistribution in the soil profile is restricted to the first layers. In contrast, in the deeper layer, especially up to 100 cm, the decrease of soil moisture continues, reaching low values due to the transpiration of the trees.

SWS in plot 24 (*M. willdenowii*) stands out as the consistently highest values throughout the evaluation period, ranging from 359.0 mm to 631.0 mm. Plot 42 (*B. salicifolius*) was the second most prominent in the SWS (ranging from 300 to 501.4 mm). The lowest values were observed in plots 45 and

30, represented by *C. langsdorffii* and *M. willdenowii*, with variations from 246.7 mm to 398.5 mm and 228.2 mm to 416.0 mm, respectively. In the other plots, the SWS varied from 279.7 mm to 476.5 mm, from 304.1 mm to 513.2 mm, and from 288.2 mm to 483.2 mm at the respective plots 60, 62, and 6, represented by *X. brasiliensis*. The variation in SWS between monitoring plots is related to differences in topographic orientation, preferential flows, soil hydrological properties, root distribution (Junqueira Júnior et al., 2017; Rodrigues et al., 2020), and spatial distribution of EP (Rodrigues et al., 2020).

Figure 7 shows the evapotranspiration (ET_{wb}) behavior in the species studied at SSMF. ET_{wb} was the highest for most trees in June, ranging from 28.0 mm to 82.4 mm. However, these values did not exceed those observed in September for *M. willdenowii*, *C. langsdorffii*, and *X. brasiliensis* (P30, P45, and P60, respectively) and in August for *M. willdenowii* and *X. brasiliensis* (P24 and P62, respectively). April and July were the critical months for ET_{wb} for almost all species evaluated, with maximum values of 47.5 mm in April and 45.8 mm in July, *M. willdenowii* and *X. brasiliensis*, respectively.

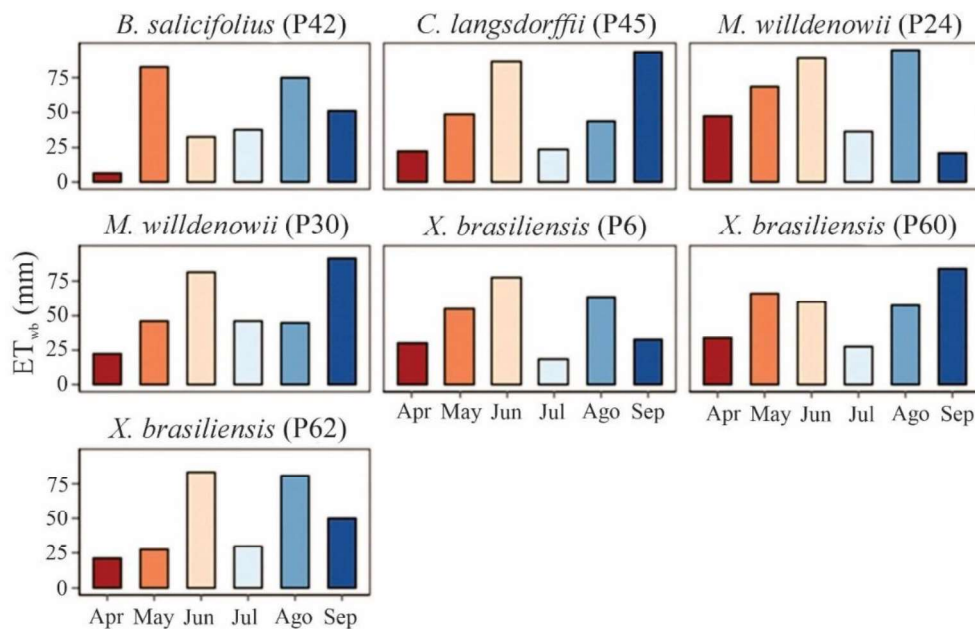


Figure 7. ET_{wb} calculated for the species evaluated in SSMF.

The temporal variability of ET_{wb} is associated with the meteorological conditions within and over the forest canopy, SWS dynamics, and EP pattern at each monitoring plot. September was the month in which high ET_{wb} values were observed for *M. willdenowii* (P30), *C. Langsdorffii* (P45), and *X. brasiliensis* (P60). One of the factors that explain this behavior is that, in the transition period from dry to rainy season, when the ecosystem is recovering, GP tends to be higher, implying an increase in soil water availability and in solar radiation at the end of winter (Deng et al., 2024). On the other hand, conditions of greater cloudiness and reduced solar radiation in autumn (from the end of April) may have contributed to the reduction in ET_{wb} in April. Yet, July presented low ET_{wb} because of decreased soil moisture, increased atmospheric demand, and the absence of EP. It should also be noted that half of the SSMF trees lose their leaves during this period, reducing physiological activities.

Plant morphology and stomatal regulation are crucial in ET_{wb} variability (Wei et al., 2017). In Figure 7, it is observed that the highest ET_{wb} values were obtained at *M. willdenowii* (P24 and P30) and *C. langsdorffii* (P45), reaching 94.5 mm in August in P24, 91.2 mm in September in P30 and 93.5 mm in September in P45. The minimum monthly values observed for these species were approximately 22 mm. These species stand out for dense canopy plots with lower plant density and higher SWS. During this time of year, phenological characteristics of species like *C. langsdorffii*, are marked by leaf shedding at higher rates (Dias, 1995; Oliveira-Filho, 1994), while the leafing phase peaks from the second half of August to the end of October (Pedroni et al., 2002), directly influencing the high ET rates observed. Moreover, the effects of the El-Niño phenomenon (INPE, 2024) in the studied period may have boosted evapotranspiration in August and September by increased temperature and reduced GP, leading to a drier period than normal.

X. brasiliensis, with a conical crown, presented a maximum ET_{wb} of 83.8 mm in September at plot 60, and 83.1 mm and 77.4 mm at plots 62 and 6 in June, respectively. The minimum values were 27.6 mm in June at point 60, 21.1 mm at point 62 in June, and 19.2 mm in April at point 6. In the case of these species, leafing and shedding do not follow seasonal patterns (Dias, 1995; Oliveira-Filho, 1994). Although

the leafing curve shows some reductions in the dry season and the beginning of the rainfall season, most individuals re-leaf throughout the year, and only a few exhibits total or partial leaf shedding.

B. salicifolius (P42), under greater tree density, presented the lowest monthly ET_{wb} , 6.5 mm in April, and reached a maximum of 82.4 mm in the evaluation period. In these cases, it is possible that the combined effect of climate and soil moisture and the ecological factors of the forest, such as competition for water and light, are mainly responsible for the lower ET_{wb} , reducing the efficiency of these plants in the eco-physiological context.

Relationship between above-ground biomass stock (AGB), ET_{wb} and CI

The data in Table 2 refers to the AGB, CI, and ET_{wb} accumulated over the monitored period in the plots and respective species.

Table 2. AGB values and CI and ET_{wb} accumulated by tree species.

Plot	Species	AGB (kg)	CI (mm)	ET_{wb} (mm)
P42	<i>Blepharocalyx salicifolius</i> (Kunth)	14.31	44.98	284.27
P60	<i>Xylopia brasiliensis</i> (Spreng.)	14.03	20.64	328.88
P62	<i>Xylopia brasiliensis</i> (Spreng.)	15.52	54.18	293.42
P45	<i>Copaifera langsdorffii</i> (Desf.)	48.96	58.84	316.04
P30	<i>Miconia willdenowii</i> (Klotzsch ex Naudin)	20.41	58.84	331.43
P06	<i>Xylopia brasiliensis</i> (Spreng.)	54.52	88.35	278.96
P24	<i>Miconia willdenowii</i> (Klotzsch ex Naudin)	24.35	23.43	357.29

X. brasiliensis in P6 has the largest AGB stock (54.5 Kg), surpassing the other trees. However, this tree species showed lower values in plots P60 and P62, with 14.0 and 15.5 Kg, respectively. Although *X. brasiliensis* developed a conical shape, the tree in P6 displayed a balanced growth, with an expansion of the crown and a higher DBH. In contrast, the trees at P60 and P62 showed apical growth, influenced by other trees in their neighborhood, resulting in a lower DBH, i.e., an effect of the ecological interaction in the forest, affecting their development if compared to P6. A similar growth pattern was observed in the tree at P42 (*B. salicifolius*), which also had a lower AGB stock (14.3 Kg). *C. langsdorffii* at P45 showed an AGB stock like *X. brasiliensis* (49.0 kg). *M. willdenowii* at P30 and P24 presented AGB stocks of 20.4 and 24.4 Kg, respectively, showing less ecological effects on their growth.

The highest accumulated CI values were observed at P6 (88.3 mm), whereas the lowest values were observed at P60 and P24, 20.6 mm and 23.4 mm, respectively, for *X. brasiliensis*. However, the species with the highest CI values generally had the highest AGB (Table 2). This suggests a positive relationship between these variables, and a Pearson coefficient of 0.7 ($p < 0.05$) was obtained. This relationship finds background on the fact that trees with larger canopies tend to have larger volumes in tropical forests, which leads to higher AGB.

The correlation between AGB and ET_{wb} was negative (-0.25) and nonsignificant. This behavior occurred because ET_{wb} was obtained for the dry period, when trees are subjected to water stress, with a reduced water supply (source) and reduced solar radiation, which led the trees into a deciduous behavior in this period. Thus, there is a period of dormancy for the trees, implicating nongrowth. The highest accumulated ET_{wb} was observed to *M. willdenowii*, with 331.4 mm (on average, 55.2 mm/month) and 357.3 mm (59.55 mm/month) at P30 and P24, respectively. Accumulated values above 300 mm (50 mm/month) were observed in *X. brasiliensis* at P60 and *C. langsdorffii* at P45, 328.9 mm (54.8 mm/month) and 316.0 mm (52.7 mm/month), respectively. The other species presented lower values, *B. salicifolius* at P42 with 284.27 mm (47.4 mm/month) and *X. brasiliensis* at P62 and P6 plots, with 293.42 mm (48.9 mm/month) and 278.96 mm (46.5 mm/month), respectively.

Relationships between soil moisture, distribution of root biomass in the soil profile and saturated soil hydraulic conductivity

An analysis based on the Pearson coefficient (Figure 8) was conducted to evaluate the relationship between soil moisture, root biomass distribution (kg m^{-3}), and soil-saturated hydraulic conductivity in the soil profile. Most of the results show a strong negative correlation between soil moisture and root biomass distribution in the profile, indicating that soil moisture tends to be inversely proportional to the concentration of root biomass. The exceptions were P42 and P60, which showed a moderate and positive correlation, and P6, which showed a nonsignificant correlation.

In the surface and intermediate soil layers (PF1: 0 – 40 cm; PF2: 40 – 100 cm), negative correlations are observed for P45 (*Copaifera*), P30 (*Miconia*) and P24 (*Miconia*) due to a continuous increase in soil moisture from the surface to the deeper layer (Figure 9b), i.e., a higher water flow in-depth, and, consequently, lower average soil moisture in the shallow layers. Besides, these layers have a higher concentration of fine roots, decaying in depth and generating a continuous reduction in biomass in the soil profile (Figure 3). The correlations decrease in the PF3 profile, where soil moisture and root concentration are lower. For P24, an increase in soil moisture was observed between PF2 and PF3, with a reduction in the concentration of roots (Figure 8). For P42 and P60 (*Blepharocalyx* and *Xylopita*, respectively), the root profile showed a lower concentration of fine roots in PF1 and PF2 than in the other plots, meaning a positive correlation between soil moisture and root mass. P6 (*Xylopia*) did not show important correlations between soil moisture and root mass, probably due to little variability in the soil profile. These contrasting correlations between soil moisture and root biomass may be due to the spatial distribution of soil and trees' characteristics. Positive correlations indicate that roots grow in locations and depths with more water availability. However, wetter locations donot ever translate into greater root uptake since this water can be strongly bound to the soil (high microporosity) (Demir et al., 2024). Trees' water demand was lower in these plots (P42 and P60) as they have the lowest AGB, although *X. brasilienses* (P60) had the second largest ET_{wb} (328.88 mm). This indicates that water availability in the shallow layers was enough to keep

evapotranspiration in the dry season since root biomass accumulation in PF1 and PF2 was lower for these locations. On the other hand, negative correlations may be due to water uptake difficulties. Trees with higher AGB demand more water to fulfill their physiological needs (e.g., in P45 and P06) and may seek it in deeper layers when water storage declines in the shallower ones (Demir et al., 2024). This is likely to happen in locations with negative correlations because of their largest root biomass accumulation, even with depth (e.g., P45 and P24), enabling them to shift the uptake to deeper layers in dry conditions (Agee et al., 2021).

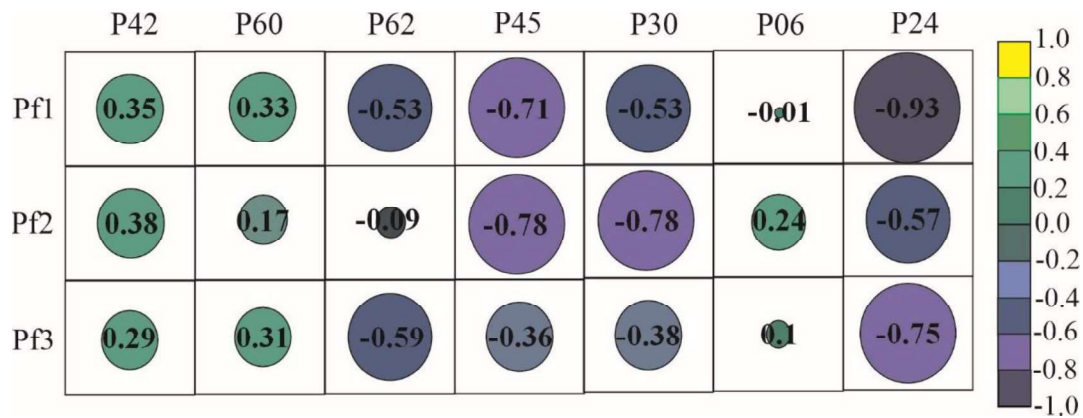


Figure 8. Correlation matrix between root biomass (BR) and soil moisture in different layers.

Figure 9 illustrates the relationship between the temporal average soil moisture and saturated soil hydraulic conductivity throughout the soil profile. Considering all monitoring points, soil moisture was consistently lower in the surface layer (0 – 20 cm), more variable in the intermediate layers (20 – 140 cm), and higher in the deeper layers (140 – 180 cm). The Ksat was higher in the surface layers, with a reduction in depth. In surface layers (Pf1) is nearer the plots are *B. salicifolius* (P42) and *X. brasiliensis* (60 and 62). The Ksat showed the highest value in the 20 – 40 cm layer (124.45 cm h⁻¹), and the average Ksat in the profile was 52.73 cm h⁻¹, higher than the other profiles. Pf2 is in the neighborhood of *C. langsdorffii* (P45) and *M. willdenowii* (P30), and the highest Ksat was obtained in the 0 – 20 cm layer (110.46 cm h⁻¹), with an average in the soil profile of 33.37 cm h⁻¹, lower than in the other profiles. Pf3 is close to the *X. brasiliensis* (P6) and *M. willdenowii* (P24), and Ksat showed the highest value in the 40 – 60 cm layer (97.02 cm h⁻¹),

and the profile average was 43.27 cm h^{-1} . Some aspects of the relationship between soil moisture and Ksat are relevant. Soil moisture tends to increase in depth. In contrast, Ksat tends to decrease, especially from a depth of 60 cm, constraining the water movement in the soil profile. Increased soil moisture at intermediate depths (40 – 60 cm) is explained by the reduction in Ksat in these layers (Figure 9).

The Ksat behavior along the soil profile can be attributed to the distribution of root biomass, the orientation, and changes in the soil structure in depth resulting from the decomposition of organic matter formed from the litter. Its concentration may vary depending on the forest species. Organic matter provides greater aggregation and cohesion of soil particles, especially in the surface layers, reducing soil density and increasing infiltration (Barbosa et al., 2019; Junqueira Júnior et al., 2017).

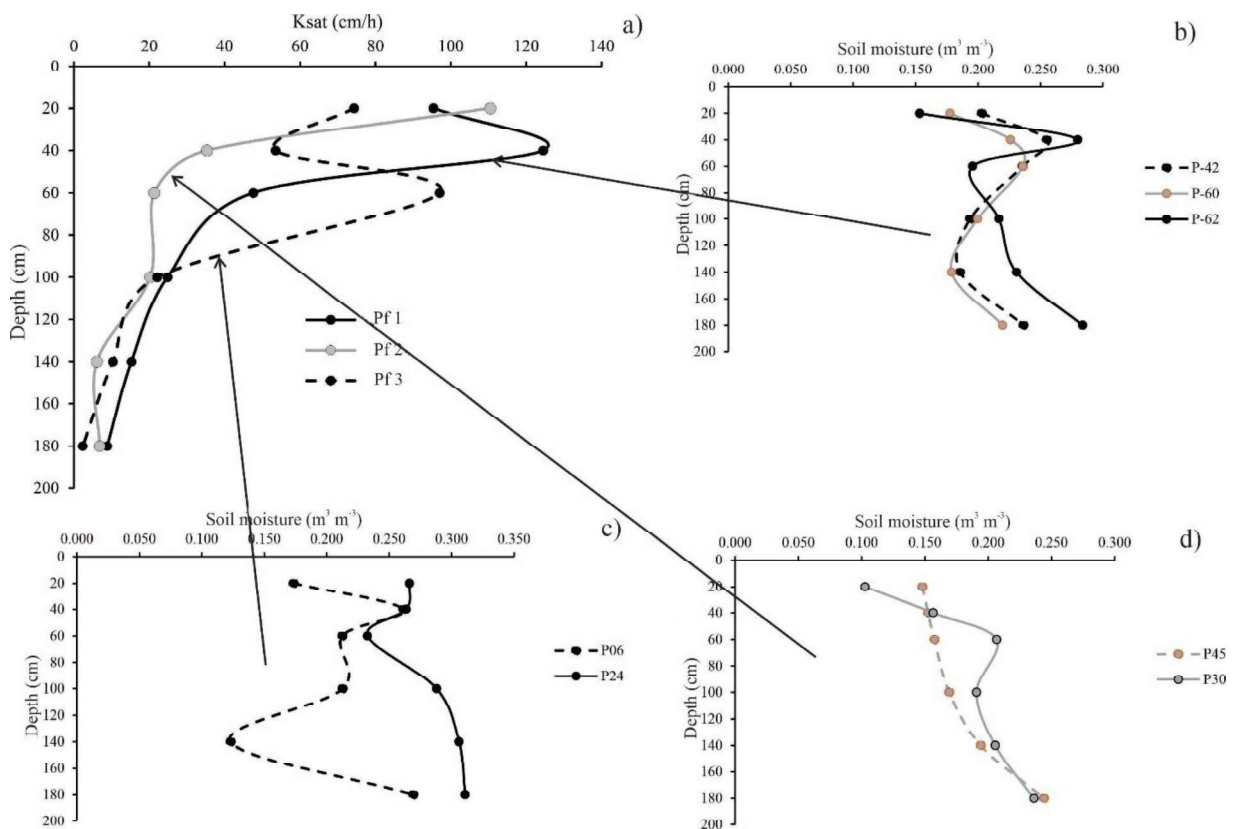


Figure 9. Ksat and temporal average soil moisture behavior in the soil profile in SSMF.

Spatial and temporal variability of soil moisture/SWS in the soil profile at SSMF

The coefficient of variation (CV) was examined to analyze the behavior of average soil moisture in time (CV_t) and space (CV_s) in the soil profile, as illustrated in Figure 10a. The CV_t showed a greater variation at depths of 20, 40, and 140 cm, with percentages of 30.4%, 23.8%, and 27.9%. The CVs presented a moderate variation between depths, which tends to stabilize with depth, ranging from 14.9% to 19.0%, with the highest percentage observed at a depth of 20 cm.

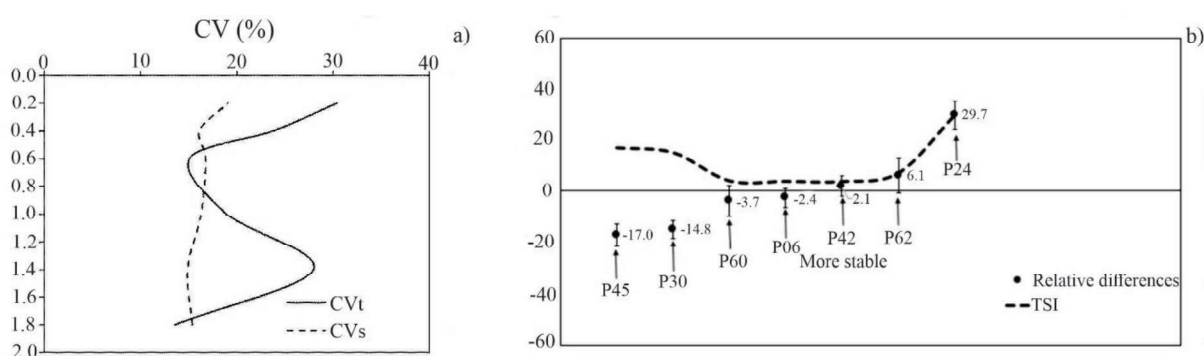


Figure 10. Distribution of temporal and spatial stability indicators of soil moisture (a) and average relative difference and Temporal Stability Index (TSI) (b) considering the monitoring plots in SSMF between April and September/2023.

The greatest variability, mainly temporal at depths of 20 cm and 40 cm, is attributed to the concentration of root biomass, as fine roots are responsible for water uptake, and the greater water infiltration in the surface layers as a result of the litter decomposition, which creates optimal conditions for infiltration. Junqueira Júnior et al. (2017) observed significant changes in soil moisture stability up to 40 cm deep in the SSMF. They attributed this greater dynamic to the direct influence of vegetation and climate, i.e., EP and ET, which affect the shallow layers more. The replenishment – consumption – replenishment process can also explain the significant changes (Xiao et al., 2014) since some rainfall events happened during the dry season (Figure 5).

About the depth of 140 cm, one of the likely factors for variation in soil moisture is the soil attributes, with emphasis on soil bulk density, which has shown an increasing trend with increasing depth, as found by Mantovani et al. (2024) in the same study area. Deeper layers showed a lower root biomass compared to the other layers. Carranza et al. (2021) stated that preferential flows produced by biological activity, such as thick roots of dead plants or animals, are also drivers of soil moisture variability in deep layers. This observation converges on the difficulty of water uptake from shallow layers with lower soil moisture (negative correlation with root biomass). Preferential flows drain water quickly from the shallow layers to deeper ones, decreasing water availability and forcing trees to invest in root growth.

In the Temporal Stability Index (TSI) of soil moisture analysis, P42 (*B. salicifolius*) presented the greatest stability, indicating less variability in the soil water storage in the dry period (Figure 10b). Moderate stability was observed at P60, P6, and P62, and the lowest stability was observed at P45, P30, and P24, showing significant fluctuations in soil water storage in the dry period. In P42, we observed a moderate to weak correlation between soil moisture and root biomass in the soil profile and one of the lowest ET (284.3 mm), which helps explain this greater temporal stability. Based on these aspects, the linked species (*B. salicifolius*) seems to be better adapted to the environment of the SSMF.

According to Mello et al. (2019) and Terra et al. (2018), forest structure is an essential factor in explaining the spatial variability of soil moisture. Rodrigues et al. (2020) observed in the same study site that the flat areas, which cover P42, P62, and P46, are linked to higher SWS, while undulated areas (P45 and P30) are related to lower SWS.

Our results shed light on the water balance in an SSMF remnant, focusing on evapotranspiration and soil moisture and their relationships with native forest aspects (root and aerial biomass). We could observe a significant relationship between soil moisture and evapotranspiration with root and above-ground mass, meaning a driver neglected in soil moisture models. In addition, soil moisture showed relationships with Ksat, which, despite being less significant than the root mass, also helps explain SWS in the soil profile.

Conclusions

In this study, the individual water balance for representative species of the Atlantic Forest was carried out at seven plots within an SSMF remnant, covering four species: *Blefarocalice salicifolius* (Kunth), *Miconia argyrophylla* DC, *Xylopia brasiliensis* Spreng. and *Copaifera langsdorffii* Desf.. The following conclusions were reached:

a) The rainfall pattern influenced the ET, EP, and CI behavior in the monitored species during the dry period of 2023. This fact is associated with the canopy architecture, canopy density, and morphological characteristics of the species.

b) The highest values of GR, EP and CI recorded in April, August and September reflect the seasonal variability of rainfall, especially in the species *X. brasiliensis* and *M. argyrophylla*, which showed greater variation in EP and lower values of SWS variation. SWS behavior varied between species, with *M. argyrophilic* presenting the highest values.

c) We observed that the higher the AGB, the higher the CI; this relationship is inverse between AGB and ET in all the species evaluated.

d) The species' roots are mainly concentrated in the surface layers (< 60 cm), associated with the availability of nutrients and water, with a non-uniform distribution in the soil profile.

e) The concentration of root biomass throughout the soil profile contributes to the variability of soil moisture, especially in the first layers; we observed an inverse relationship between soil moisture and root biomass concentration, indicating that higher root concentration results in lower soil moisture in layers < 100 cm.

f) The plots with *Blefarocalice salicifolius* and *Xylopia brasiliensis* showed a trend to greater SWS stability in the soil profile; such information can be used to establish a recovery planning for areas similar to SSMF (semi-deciduous forest in a Latosol under well-defined seasons of the year).

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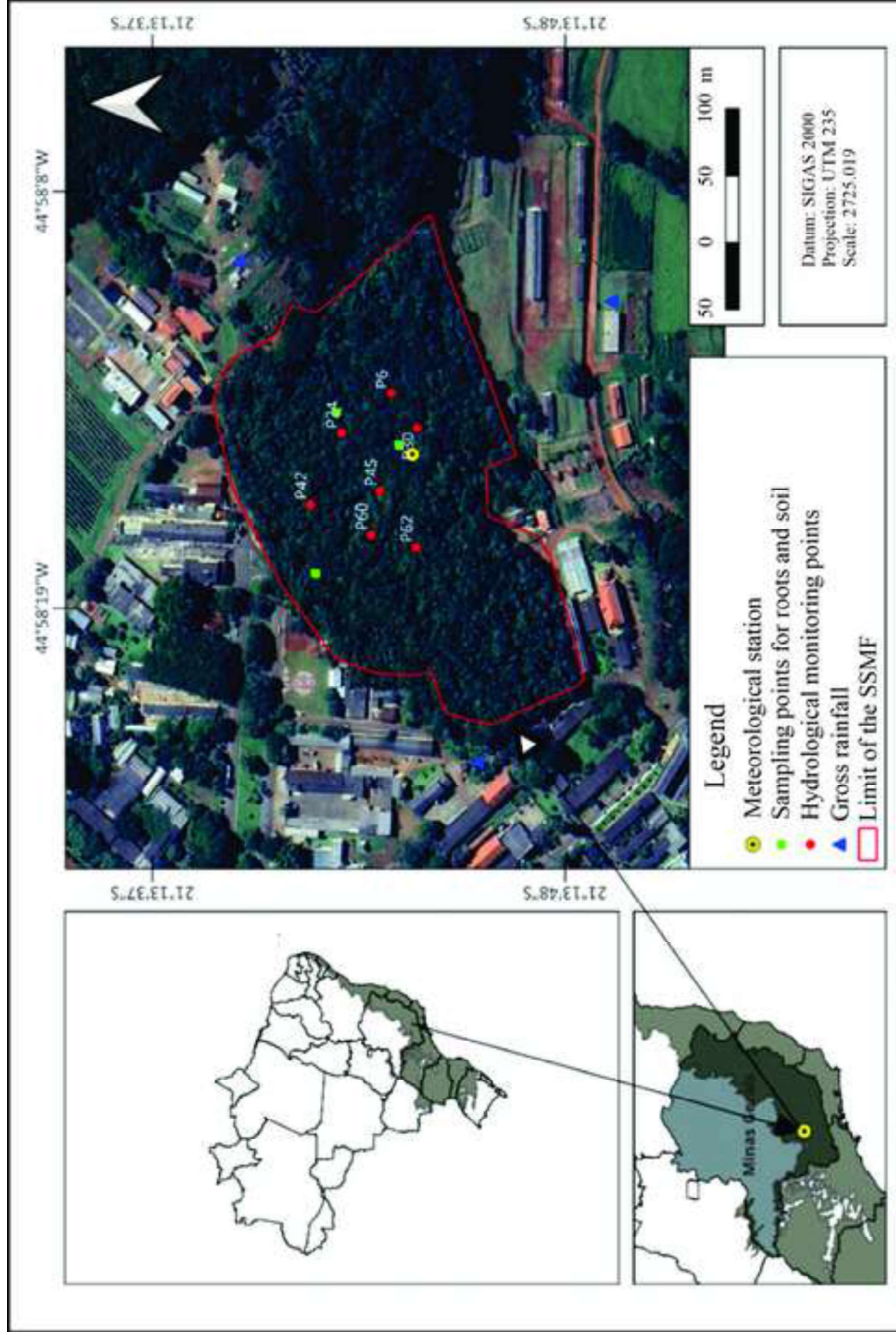


Figure 1

Figure 2



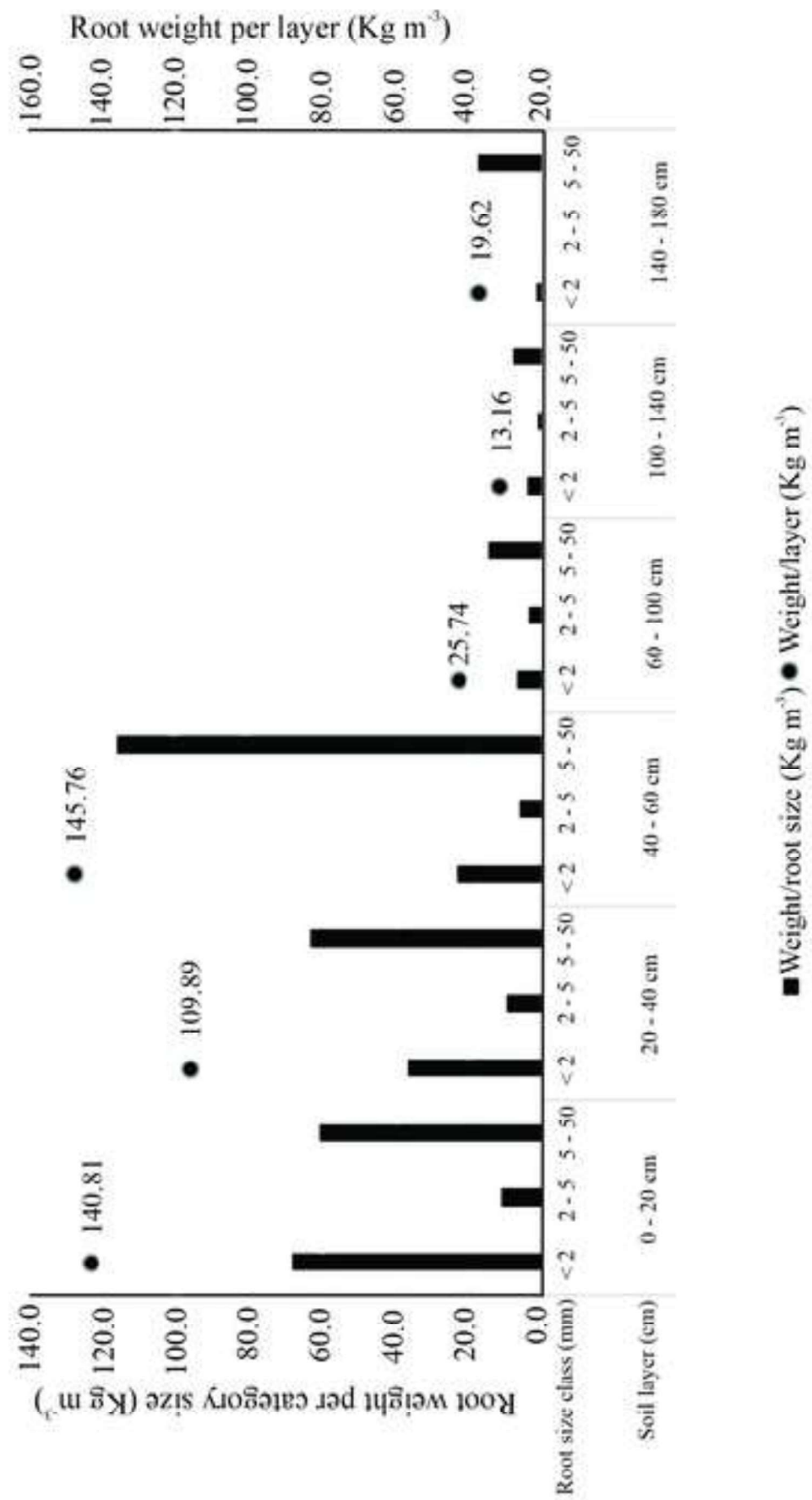
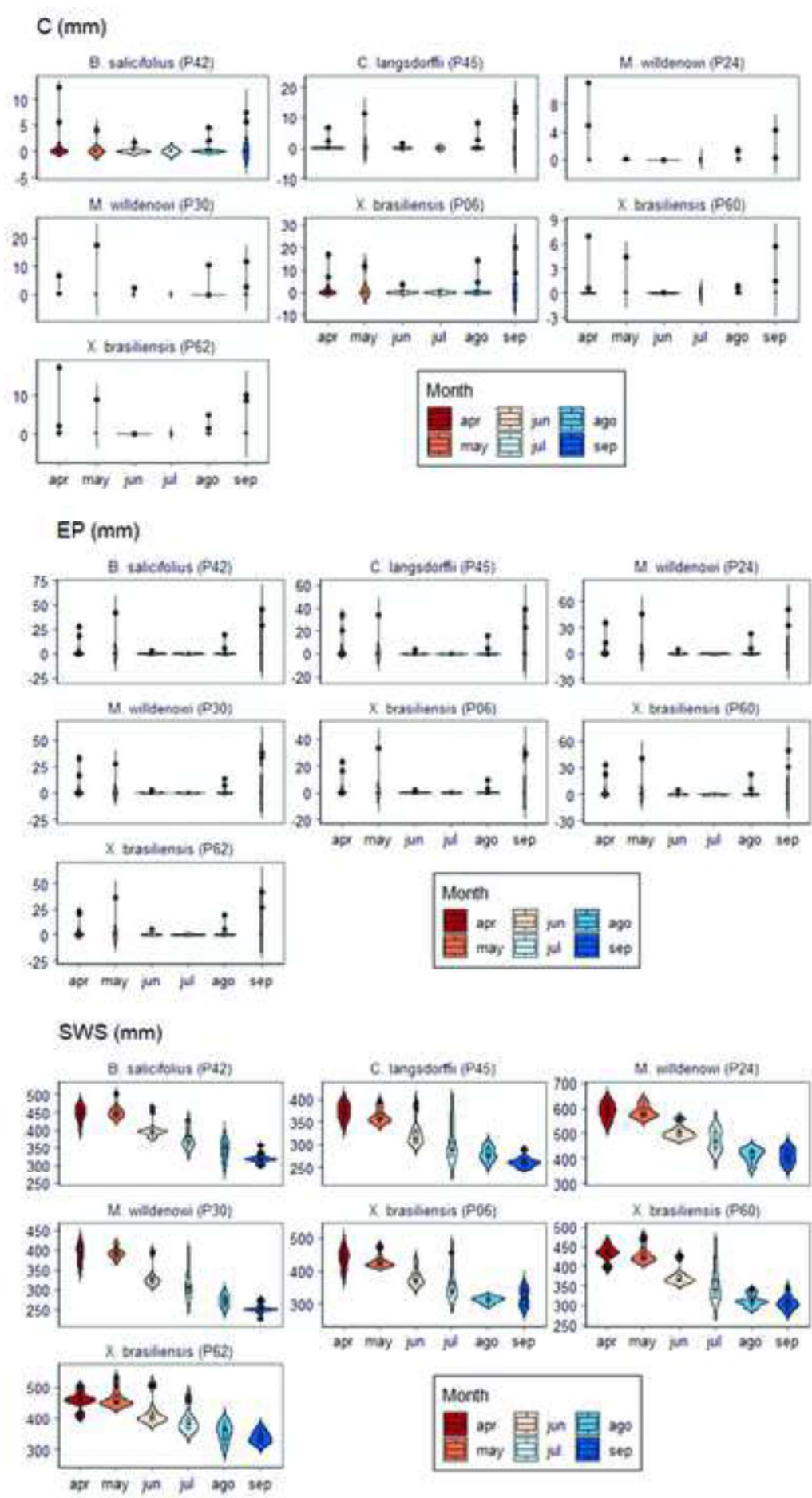


Figure 3



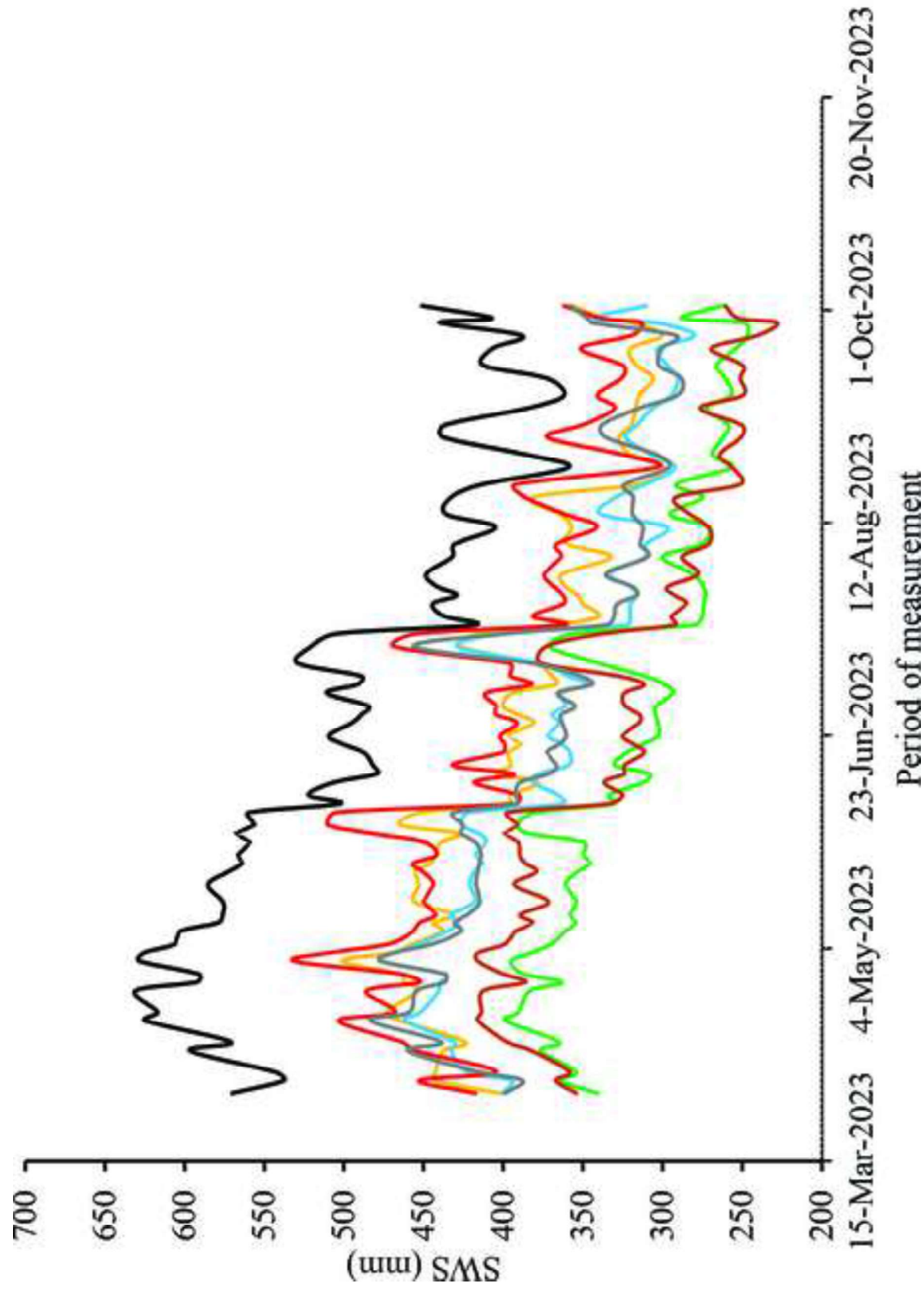
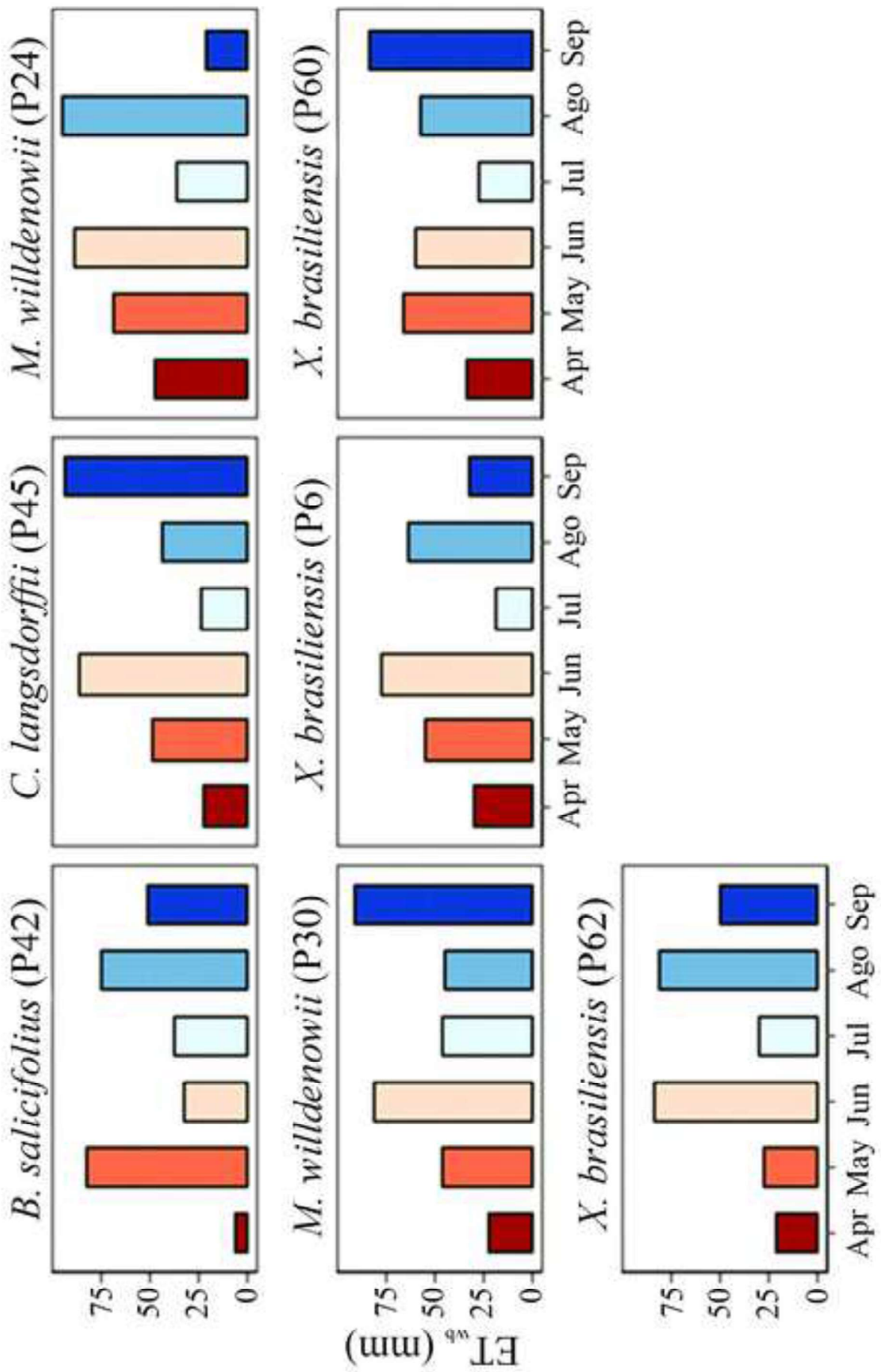
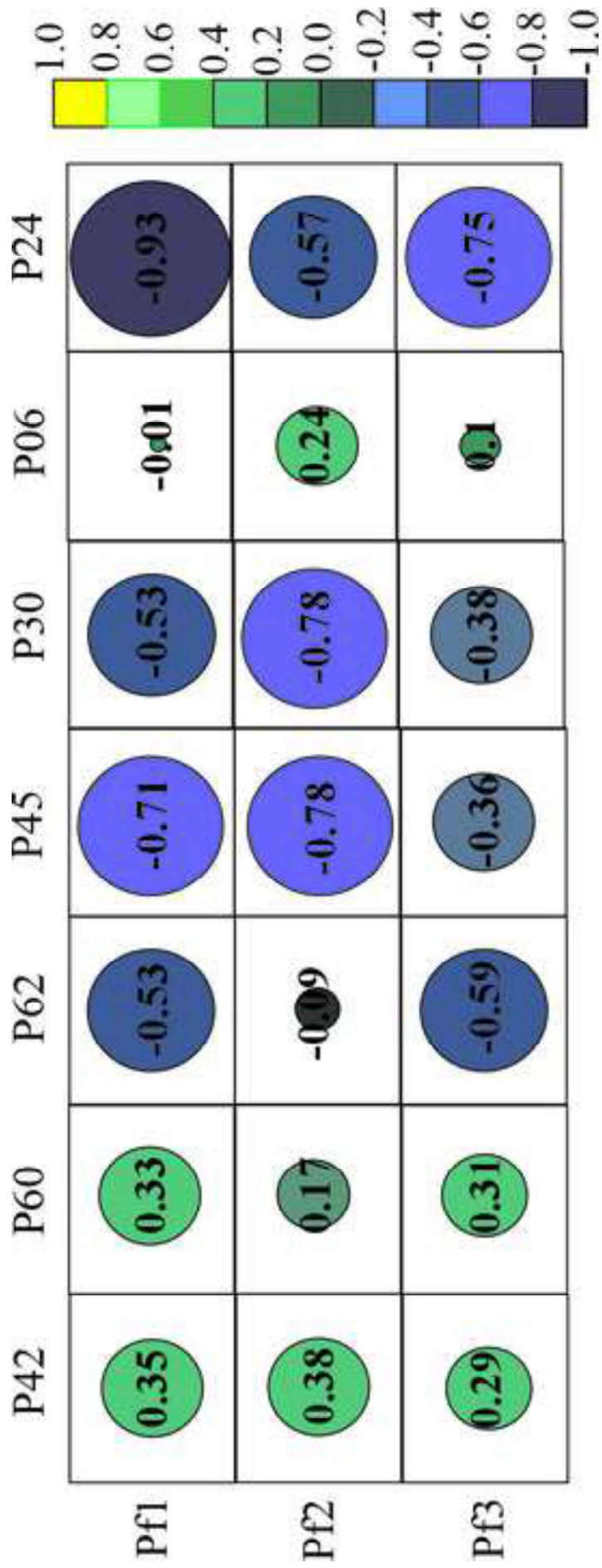
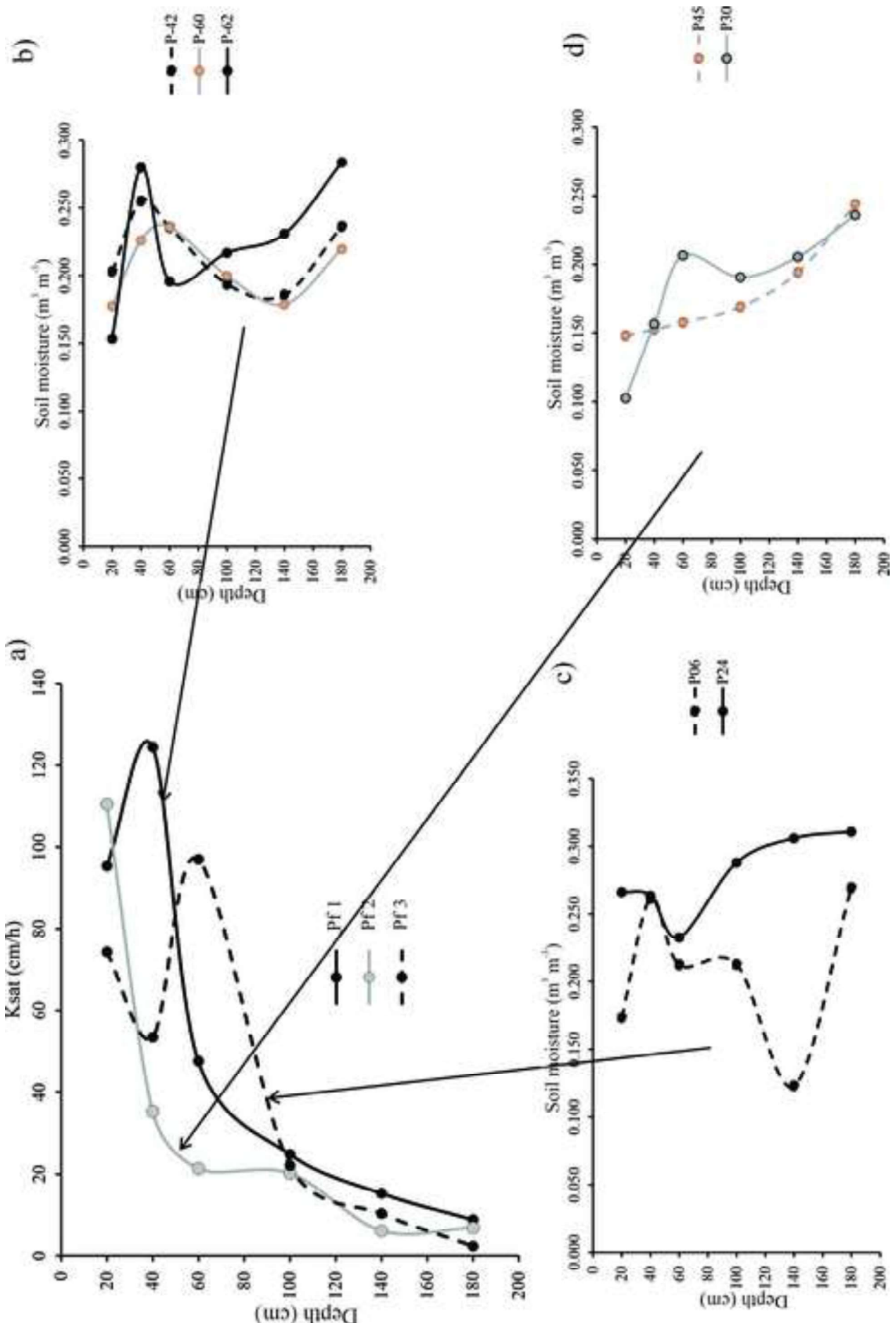


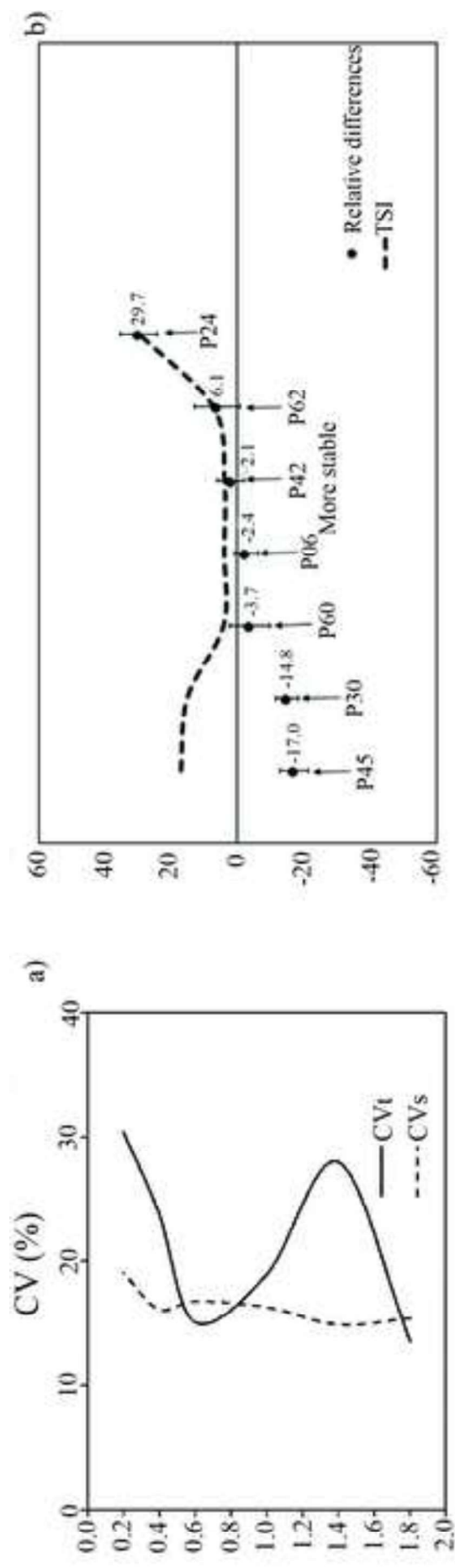
Figure 6

Figure 7









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This manuscript wasn't previously submitted to another journal, and the authors are aware of and agree with this submission to the Ecohydrology and Hydrobiology Journal.



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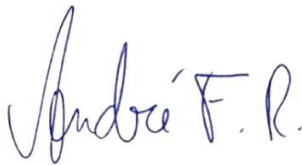
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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Highlights

- Atlantic Forest has faced severe droughts in the last decades affecting tree growth
- Some trees have shown different water balance behaviors
- Soil moisture in deeper layers showed little effect on tree water uptake in dry season
- Root mass and distribution have a crucial role in soil moisture and ET dynamics